

Chapter 6: Measuring Space — Perimeter and Area

Complete NCERT Exercise-by-Exercise Answer Key & Step-by-Step Solutions

EXERCISE SET 6.1 — Circumference & Arc Length

Question 1

The perimeter of a circle is 44 cm. What is its radius?

Solution:

Given circumference $C = 44 \text{ cm}$. Formula: $C = 2\pi r$.

$$2 \times \frac{22}{7} \times r = 44 \implies \frac{44}{7} r = 44 \implies r = 7 \text{ cm}.$$

Answer: The radius of the circle is **7 cm**.

Question 2

Calculate, correct to 3 significant figures, the circumference of a circle with:

- (i) radius 7 cm (ii) radius 10 cm (iii) radius 12 cm

Solution:

$$(i) \text{ } r = 7 \text{ cm} \implies C = 2 \times \frac{22}{7} \times 7 = \mathbf{44.0 \text{ cm}}.$$

$$(ii) \text{ } r = 10 \text{ cm} \implies C = 2 \times \frac{22}{7} \times 10 = \frac{440}{7} \approx \mathbf{62.9 \text{ cm}}.$$

$$(iii) \text{ } r = 12 \text{ cm} \implies C = 2 \times \frac{22}{7} \times 12 = \frac{528}{7} \approx \mathbf{75.4 \text{ cm}}.$$

Question 3

Calculate the length of the arc of a circle if:

- (i) the radius is 3.5 cm and the angle at the centre is 60°
(ii) the radius is 6.3 m and the angle at the centre is 120°

Solution: Arc length formula $l = 2\pi r \times \frac{\theta}{360^\circ}$.

$$(i) \text{ } l = 2 \times \frac{22}{7} \times 3.5 \times \frac{60}{360} = 22 \times 0.5 \times \frac{1}{6} = \frac{11}{3} \approx \mathbf{3.67 \text{ cm}}.$$

$$(ii) \text{ } l = 2 \times \frac{22}{7} \times 6.3 \times \frac{120}{360} = 44 \times 0.9 \times \frac{1}{3} = \mathbf{13.2 \text{ m}}.$$

Question 4

Find the perimeter of a sector (the curved portion as well as the two straight portions) of a circle of radius 14 cm and sector angle 75° .

Solution:

$$\text{Arc length } l = 2\pi r \times \frac{\theta}{360^\circ} = 2 \times \frac{22}{7} \times 14 \times \frac{75}{360} = 88 \times \frac{5}{24} = \frac{55}{3} \approx 18.33 \text{ cm}.$$

$$\text{Perimeter of sector } s = l + 2r = 18.33 + 2(14) = 18.33 + 28 = \mathbf{46.33 \text{ cm}} \text{ (or } \frac{139}{3} \text{ cm)}.$$

Question 5

Find the perimeters of composite shapes (quarter, half, or three-quarter circular arcs):

Solutions for key figures (Fig. 6.14i–ix):

- **(i) Track shape (80m × 60m):** Straight sides $= 2 \times 80 = 160 \text{ m}$. Two semicircles of radius 30 m form 1 full circle of circumference $2\pi(30) = 60 \times \frac{22}{7} = 188.57 \text{ m}$. Total $= 160 + 188.57 = 348.57 \text{ m}$.
- **(ii) Arch (width 12cm, height 8cm):** Outer arc $(\pi \times 6 = 18.86 \text{ cm})$ + inner arc $(\pi \times 4 = 12.57 \text{ cm})$ + two base steps $(2 \times 2 = 4 \text{ cm}) = 35.43 \text{ cm}$.
- **(iii) 4-lobed square (10cm side):** 4 semicircles of radius $5 \text{ cm} = 2 \text{ full circles} = 2 \times 2\pi(5) = 20\pi = 62.86 \text{ cm}$.
- **(vii) Right triangle (6cm, 8cm, 10cm):** 3 semicircular arcs on sides $= \pi(3) + \pi(4) + \pi(5) = 12\pi = 37.71 \text{ cm}$.
- **(ix) Semicircular ends (10cm dia, 10cm straight):** $2(10) + 2\pi(5) = 20 + 10\pi = 51.43 \text{ cm}$.

Question 6

If the diameter of a car tyre is 56 cm, then:

- How far does the car need to travel for the tyre to complete one revolution?
- How many revolutions does the tyre make if the car travels 10 km?

Solution:

- Distance in 1 revolution $= \text{Circumference } C = \pi D = \frac{22}{7} \times 56 = 176 \text{ cm} = 1.76 \text{ m}$.
- Total distance $= 10 \text{ km} = 10,000 \text{ m}$.
Number of revolutions $= \frac{10000}{1.76} \approx 5682 \text{ revolutions}$.

Question 7

Find the total perimeter of all the petals in each flower figure:

- Square of side 14 cm with midpoint arcs
- Regular hexagon of side 42 cm with vertex arcs

Solution:

- Square of side 14 cm : Arcs centered at midpoints ($r = 7 \text{ cm}$). 4 semicircles form 4 petals. Perimeter $= 4 \times (\pi \times 7) = 4 \times 22 = 88 \text{ cm}$.
- Hexagon of side 42 cm : Arcs centered at vertices ($r = 21 \text{ cm}$). 6 petals formed by 12 arcs of 60° each $= 2 \text{ full circles} = 2 \times (2\pi \times 21) = 84\pi = 264 \text{ cm}$.

Question 8

The ratio of the perimeters of two circles is 5:4. What is the ratio of their radii?

Solution:

$$\text{Ratio of perimeters } = \frac{2\pi r_1}{2\pi r_2} = \frac{r_1}{r_2} = \frac{5}{4}$$

Answer: The ratio of their radii is 5:4.

EXERCISE SET 6.2 — Area of Polygons & Heron's Formula

Question 1

Find the area of triangle ADE in Fig. 6.31 (base DE = 10 cm, height = 8 cm).

Solution:

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 10 \times 8 = 40 \text{ cm}^2.$$

Question 2

The parallel sides of a trapezium are 40 cm and 20 cm. If its non-parallel sides are both equal, each being 26 cm, find the area of the trapezium.

Solution:

$$\text{Base projection } x = \frac{40 - 20}{2} = 10 \text{ cm}.$$

$$\text{Height } h = \sqrt{26^2 - 10^2} = \sqrt{676 - 100} = \sqrt{576} = 24 \text{ cm}.$$

$$\text{Area} = \frac{1}{2}(a + b)h = \frac{1}{2}(40 + 20) \times 24 = 30 \times 24 = 720 \text{ cm}^2.$$

Question 3

Find the area of a triangle, given that its sides are 8 cm and 11 cm long, and its perimeter is 32 cm.

Solution:

$$\text{Third side } c = 32 - (8 + 11) = 13 \text{ cm}. \text{ Semi-perimeter } s = \frac{32}{2} = 16 \text{ cm}.$$

$$\text{Heron's Formula: } \text{Area} = \sqrt{16(16-8)(16-11)(16-13)} = \sqrt{16 \times 8 \times 5 \times 3} = \sqrt{1920} = 8\sqrt{30} \approx 43.82 \text{ cm}^2.$$

Question 4

The sides of a triangular plot are in the ratio 3 : 5 : 7; its perimeter is 300 m. Find its area.

Solution:

$$\text{Let sides be } 3x, 5x, 7x. \quad 3x + 5x + 7x = 300 \implies 15x = 300 \implies x = 20 \text{ m}.$$

$$\text{Sides: } a = 60 \text{ m}, b = 100 \text{ m}, c = 140 \text{ m}. \quad s = 150 \text{ m}.$$

$$\text{Area} = \sqrt{150(150-60)(150-100)(150-140)} = \sqrt{150 \times 90 \times 50 \times 10} = \sqrt{6750000} = 1500\sqrt{3} \approx 2598.08 \text{ m}^2.$$

Question 5

One diagonal of a rhombus is twice as long as the other diagonal. If the rhombus has area 128 cm^2 , find the length of the shorter diagonal.

Solution:

$$\text{Let shorter diagonal be } d_1, \text{ then } d_2 = 2d_1.$$

$$\text{Area} = \frac{1}{2} d_1 d_2 = \frac{1}{2} d_1 (2d_1) = d_1^2 = 128 \implies d_1 = \sqrt{128} = 8\sqrt{2} \approx 11.31 \text{ cm}.$$

Question 6

ABCD is a parallelogram. P and Q are any two points on side AB. What can you say about the ratio $\text{area}(\Delta PCD) : \text{area}(\Delta QCD)$?

Solution:

Both ΔPCD and ΔQCD lie on the same base CD and between the same parallel lines $AB \parallel CD$. Thus they have identical height h .

$\text{Area}(\Delta PCD) = \frac{1}{2} CD \cdot h = \text{Area}(\Delta QCD)$. Ratio is **1 : 1**.

Question 7

O is any point on the diagonal PR of a parallelogram PQRS. Prove that the areas of triangles PSO and PQO are equal.

Solution:

Diagonal PR divides $PQRS$ into two equal areas: $\text{Area}(\Delta PQR) = \text{Area}(\Delta PSR)$.

Similarly, RO divides ΔSQR into equal areas: $\text{Area}(\Delta PRO) = \text{Area}(\Delta SRO)$.

Subtracting gives: $\text{Area}(\Delta PSO) = \text{Area}(\Delta PQO)$. **(Proved)**

Question 8

If the mid-points of the sides of a 4-gon (quadrilateral) are joined in order, prove that the area of the parallelogram thus formed will be half of the area of the given 4-gon.

Solution:

By Varignon's Theorem, joining midpoints E, F, G, H creates a parallelogram.

The 4 corner triangles removed have areas equal to $\frac{1}{4}$ of the corresponding diagonal triangles. Their sum equals $\frac{1}{2} \text{Area}(ABCD)$.

Hence, $\text{Area}(EFGH) = \frac{1}{2} \text{Area}(ABCD)$. **(Proved)**

Question 9

In ΔABC , the midpoint of BC is D . Median AD is drawn. P is any point on AD . Show that $\text{area}(\Delta ABP) = \text{area}(\Delta ACP)$.

Solution:

Median AD implies $\text{Area}(\Delta ABD) = \text{Area}(\Delta ACD)$.

Median PD in ΔPBC implies $\text{Area}(\Delta PBD) = \text{Area}(\Delta PCD)$.

Subtracting equations: $\text{Area}(\Delta ABP) = \text{Area}(\Delta ACP)$. **(Proved)**

Question 10

Given a square $ABCD$, let P be a point within it. Join PA, PB, PC, PD . What is the ratio of the areas of the red region ($\Delta PAB + \Delta PCD$) and the green region ($\Delta PBC + \Delta PDA$)?

Solution:

Let square side be s . $\text{Area}(\Delta PAB) + \text{Area}(\Delta PCD) = \frac{1}{2} s h_1 + \frac{1}{2} s h_2 = \frac{1}{2} s (h_1 + h_2) = \frac{1}{2} s^2$.

The remaining green region also equals $\frac{1}{2} s^2$. Ratio is **1 : 1**.

Question 11

In $\triangle ABC$, D is the midpoint of AB. P is any point on BC, and Q is a point on AB such that $CQ \parallel PD$. Prove that $\text{Area}(\triangle BPQ) = \frac{1}{2} \text{Area}(\triangle ABC)$.

Solution:

Since $CQ \parallel PD$, $\text{Area}(\triangle PDC) = \text{Area}(\triangle PDQ)$ implies $\text{Area}(\triangle BPQ) = \text{Area}(\triangle BDC)$.

Since D is the midpoint of AB, CD is a median implies $\text{Area}(\triangle BDC) = \frac{1}{2} \text{Area}(\triangle ABC)$.

Thus, $\text{Area}(\triangle BPQ) = \frac{1}{2} \text{Area}(\triangle ABC)$. (Proved)

EXERCISE SET 6.3 — Area of Circle Sectors & Segments**Question 1**

Find the area of a sector of a circle with radius 7 cm if the angle of the sector is 60° .

Solution:

$\text{Area} = \pi r^2 \times \frac{\theta}{360^\circ} = \frac{22}{7} \times 49 \times \frac{60}{360} = \frac{154}{6} = \frac{77}{3} \approx 25.67 \text{ cm}^2$.

Question 2

Find the area of a quadrant of a circle whose circumference is 44 cm.

Solution:

$2\pi r = 44$ implies $r = 7 \text{ cm}$.

$\text{Quadrant Area} = \frac{1}{4} \pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 49 = \frac{154}{4} = 38.5 \text{ cm}^2$.

Question 3

The length of the minute hand of a clock is 7 cm. Find the area swept by the minute hand in 10 minutes.

Solution:

Angle swept in 10 min = $10 \times 6^\circ = 60^\circ$.

$\text{Area} = \frac{60}{360} \times \frac{22}{7} \times 49 = \frac{77}{3} \approx 25.67 \text{ cm}^2$.

Question 4

A chord of a circle of radius 10 cm subtends 90° at the centre. Find the area of the corresponding: (i) minor sector, and (ii) major sector. (Use $\pi \approx 3.14$)

Solution:

(i) Minor sector area = $\frac{90}{360} \times 3.14 \times 100 = 0.25 \times 314 = 78.5 \text{ cm}^2$.

(ii) Major sector area = $(3.14 \times 100) - 78.5 = 314 - 78.5 = 235.5 \text{ cm}^2$.

Question 5

A chord of a circle of radius 15 cm subtends 60° at the centre. Find the areas of corresponding minor and major segments. (Use $\pi \approx 3.14$, $\sqrt{3} \approx 1.73$)

Solution:

Sector area $= \frac{60}{360} \times 3.14 \times 225 = 117.75 \text{ cm}^2$.

Triangle area $= \frac{\sqrt{3}}{4} r^2 = \frac{1.73}{4} \times 225 = 97.3125 \text{ cm}^2$.

- Minor segment $= 117.75 - 97.3125 = 20.44 \text{ cm}^2$.
- Major segment $= 706.5 - 20.44 = 686.06 \text{ cm}^2$.

Question 6

A car has two wipers (blade 28 cm, sweeps 120°). Find total area cleaned at each sweep.

Solution:

$\text{Total Area} = 2 \times \left(\frac{120}{360} \times \frac{22}{7} \times 28^2 \right) = 2 \times \frac{2464}{3} = \frac{4928}{3} \approx 1642.67 \text{ cm}^2$.

Question 7

A chord of radius r subtends 60° at centre. Show that minor segment area $= \pi r^2 (1/6 - \sqrt{3}/4\pi)$.

Solution:

Sector area $= \frac{1}{6} \pi r^2$. Equilateral triangle area $= \frac{\sqrt{3}}{4} r^2$.

$\text{Segment Area} = \frac{1}{6} \pi r^2 - \frac{\sqrt{3}}{4} r^2 = \pi r^2 \left(\frac{1}{6} - \frac{\sqrt{3}}{4\pi} \right)$. (Proved)

Question 8

Equilateral triangle inscribed in circle of radius r . Show ratio of area of triangle to circle $= 3\sqrt{3} / 4\pi \approx 0.413$.

Solution:

Triangle side $a = \sqrt{3} r$ implies $\text{Area} = \frac{\sqrt{3}}{4} (\sqrt{3} r)^2 = \frac{3\sqrt{3}}{4} r^2$. Circle area $= \pi r^2$.

$\text{Ratio} = \frac{3\sqrt{3}}{4\pi} \approx 0.413$. (Proved)

Question 9

Square inscribed in circle of radius r . Show ratio of area of square to circle $= 2 / \pi \approx 0.637$.

Solution:

Diagonal $= 2r$ implies $\text{Square side } a = \sqrt{2} r$ implies $\text{Area} = 2r^2$.

$\text{Ratio} = \frac{2r^2}{\pi r^2} = \frac{2}{\pi} \approx 0.637$. (Proved)

Question 10

Hexagon inscribed in circle of radius r . Show ratio of area $= \frac{3\sqrt{3}}{2\pi} \approx 0.827$. Why is it twice Question 8?

Solution:

Hexagon area $= 6 \times \frac{\sqrt{3}}{4} r^2 = \frac{3\sqrt{3}}{2} r^2$. Ratio $= \frac{3\sqrt{3}}{2\pi} \approx 0.827$.

It is exactly twice Question 8 because a regular hexagon inscribed in a circle is composed of 6 equilateral triangles, whereas an inscribed triangle has only 3.

END-OF-CHAPTER EXERCISES (Questions 1 to 27)**End-of-Chapter Q1**

Draw figures corresponding to area identities $(a+b)(a-b) = a^2 - b^2$ and $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$.

Solution:

- $(a+b)(a-b) = a^2 - b^2$: A large square of side a with a corner $b \times b$ removed leaves area $a^2 - b^2$, which cuts into a rectangle of length $(a+b)$ and width $(a-b)$.
- $(a+b+c)^2$: A square of side $(a+b+c)$ partitioned into 3 squares (a^2, b^2, c^2) and 6 rectangles ($2ab, 2bc, 2ca$).

End-of-Chapter Q2

An isosceles triangle has perimeter 40 cm; the equal sides are 15 cm each. Find the area of the triangle.

Solution:

Base $c = 40 - (15 + 15) = 10 \text{ cm}$. Height $h = \sqrt{15^2 - 5^2} = \sqrt{200} = 10\sqrt{2} \text{ cm}$.
 $\text{Area} = \frac{1}{2} \times 10 \times 10\sqrt{2} = 50\sqrt{2} \approx 70.71 \text{ cm}^2$.

End-of-Chapter Q3

An isosceles triangle has base 10 cm, and its area is 60 cm². What are the lengths of the equal sides?

Solution:

$\text{Area} = \frac{1}{2} b h = 60 \implies \frac{1}{2}(10)h = 60 \implies h = 12 \text{ cm}$.
 Equal side $a = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13 \text{ cm}$.

End-of-Chapter Q4

The area of a right-angled triangle is 54 sq. cm. One of its legs has length 12 cm. Find its perimeter.

Solution:

Leg $b = \frac{2 \times 54}{12} = 9 \text{ cm}$. Hypotenuse $c = \sqrt{12^2 + 9^2} = 15 \text{ cm}$.
 Perimeter $= 12 + 9 + 15 = 36 \text{ cm}$.

End-of-Chapter Q5

The sides of a triangle are in the ratio 2 : 3 : 4, and its perimeter is 45 cm. Find its area.

Solution:

Sides: $2x + 3x + 4x = 45 \implies x = 5 \implies a=10, b=15, c=20 \text{ cm}$. $s = 22.5 \text{ cm}$.

$\text{Area} = \sqrt{22.5(12.5)(7.5)(2.5)} = \frac{75\sqrt{15}}{4} \approx 46.48 \text{ cm}^2$.

End-of-Chapter Q6

The sides of a triangle have lengths 7 cm, 24 cm, 25 cm. Find the area of the triangle in two different ways.

Solution:

• **Method 1 (Right Triangle):** $7^2 + 24^2 = 625 = 25^2 \implies \text{Area} = \frac{1}{2} \times 7 \times 24 = 84 \text{ cm}^2$.

• **Method 2 (Heron's Formula):** $s = 28 \implies \text{Area} = \sqrt{28(21)(4)(1)} = \sqrt{7056} = 84 \text{ cm}^2$.

End-of-Chapter Q7

If the wheel of a bicycle has a diameter of 60 cm, find how far a cyclist will have travelled after the wheel has rotated 100 times.

Solution:

Circumference $C = \pi D = 60\pi \approx 188.57 \text{ cm} = 1.8857 \text{ m}$.

Distance in 100 rotations $= 100 \times 1.8857 = 188.57 \text{ m}$ (or $60\pi \text{ m}$).

End-of-Chapter Q8

Find the area of a quadrant of a circle whose circumference is 66 cm.

Solution:

$2\pi r = 66 \implies 2 \times \frac{22}{7} r = 66 \implies r = 10.5 \text{ cm}$.

$\text{Quadrant Area} = \frac{1}{4} \times \frac{22}{7} \times (10.5)^2 = 86.625 \text{ cm}^2$.

End-of-Chapter Q9

The wheel of a car has an outer radius of 28 cm. Calculate distance after 1 complete turn, and turns during 1 km.

Solution:

• Distance in 1 turn $= 2\pi(28) = 176 \text{ cm} = 1.76 \text{ m}$.

• Turns in $1 \text{ km} = \frac{1000 \text{ m}}{1.76 \text{ m}} \approx 568 \text{ turns}$.

End-of-Chapter Q10

Two rectangles have the same area and the same perimeter. Does this mean that they are congruent to each other?

Solution:

Let lengths be l, w . $l+w = S/2$ and $lw = A$. l, w are the unique roots of quadratic $t^2 - (S/2)t + A = 0$.

Answer: Yes, they must have identical dimensions and are therefore congruent.

End-of-Chapter Q11–13

Trapezium area derivations (using parallelograms, triangles, and 2 identical copies):

Solution:

All three geometric proofs demonstrate that $\text{Area}(\text{Trapezium}) = \frac{1}{2}(a + b)h$, where a, b are parallel sides and h is height.

End-of-Chapter Q14

Show that the area of a kite is half the product of its diagonals using algebra and geometry.

Solution:

- **Algebraic:** Diagonals d_1, d_2 intersect at 90° . $\text{Area} = \frac{1}{2} d_1 \times \frac{1}{2} d_2 (d_1 + d_2) = \frac{1}{4} d_1 d_2$.
- **Geometric:** Enclose kite in rectangle of $d_1 \times d_2$; kite area equals half the bounding rectangle.

End-of-Chapter Q15

Three problems about fitting congruent shapes (scaling factor k):

Solution:

Scaling side lengths by k scales area by k^2 .

- (i) Scale $2x \implies \text{Area } 4x$. **Yes**, 4 copies fit.
- (ii) Triangle scale $2x \implies \text{Area } 4x$. **Yes**, 4 copies fit.
- (iii) Triangle scale $3x \implies \text{Area } 9x$. **Yes**, 9 copies fit.

End-of-Chapter Q16–18

Rectangle covered by grid-packed circles conjecture:

Solution:

For N identical circles packed in a grid inside bounding rectangle: $\text{Fraction} = \frac{N \pi r^2}{(2mr)(2nr)} = \frac{\pi}{4} \approx 78.54\%$. This ratio is constant for any number of circles.

End-of-Chapter Q19

Nine identical rectangles form a large rectangle of area 72 cm^2 . Find the perimeter of each small rectangle.

Solution:

Area of 1 small rectangle $= 72 / 9 = 8 \text{ cm}^2$.

Grid alignment gives $l = 4 \text{ cm}$, $w = 2 \text{ cm}$ ($l \times w = 8$, $2l = 4w$).

Perimeter $= 2(4 + 2) = 12 \text{ cm}$.

End-of-Chapter Q20

Trisection lines from vertex in a triangle:

Solution:

Both blue and red triangles have equal base length ($1/3$ of side) and same height h . Thus $\text{Area}(\text{blue}) = \text{Area}(\text{red})$.

End-of-Chapter Q21

Quarter circle and two semicircles in square of side s . Show $\text{Area}(A) = \text{Area}(B)$.

Solution:

$$\text{Area}(A) = \text{Quarter circle} - \text{Overlap} = \frac{1}{4}\pi s^2 - O$$

$$\text{Area}(B) = 2 \times \text{Semicircle} - \text{Overlap} = 2\left(\frac{1}{8}\pi s^2\right) - O = \frac{1}{4}\pi s^2 - O$$

Thus $\text{Area}(A) = \text{Area}(B)$. **(Proved)**

End-of-Chapter Q22

4-petalled flower in square of side 2 units. Find perimeter and area.

Solution:

$$\text{Area} = 2 \times (4 \times \text{semicircles}) - \text{Square} = 2\pi(1)^2 - 2^2 = 2\pi - 4 \approx 2.28 \text{ sq units}.$$

$$\text{Perimeter} = 4 \times (\pi \times 1) = 4\pi \approx 12.57 \text{ units}.$$

End-of-Chapter Q23

Concentric circles with tangent chord $BC = l$. Show green ring area $= \frac{1}{4}\pi l^2$.

Solution:

By Baudhāyana-Pythagoras theorem: $R^2 - r^2 = (l/2)^2 = \frac{l^2}{4}$.

$$\text{Ring Area} = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \frac{1}{4}\pi l^2. \text{ (Proved)}$$

End-of-Chapter Q24

Lunes of Hippocrates: Right triangle sides a, b, c . Show $\text{Area}(A) + \text{Area}(B) = \text{Area}(C)$.

Solution:

$$\text{Area}(A) + \text{Area}(B) = \frac{\pi a^2}{8} + \frac{\pi b^2}{8} = \frac{\pi}{8}(a^2 + b^2) = \frac{\pi c^2}{8} = \text{Area}(C). \text{ (Proved)}$$

End-of-Chapter Q25

Two congruent circles of radius r passing through each other's centers. Find area of enclosed region.

Solution:

$$\text{Area} = 2 \times \text{Sector}(120^\circ) - \text{Rhombus} = 2\left(\frac{1}{3}\pi r^2\right) - \frac{\sqrt{3}}{2}r^2 = r^2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) \approx 1.228 r^2.$$

End-of-Chapter Q26

Three triangles within rectangle (A, B, C) . Show area of rectangle $= 2(A+C)(B+C) / C$.

Solution:

$$\text{Derived via geometric similarity ratios of altitude partitions: } \text{Area} = \frac{2(A+C)(B+C)}{C}. \text{ (Proved)}$$

End-of-Chapter Q27

Quarter circle, semicircle, and triangle in Fig. 6.55.

Solution:

Both shaded regions E and F equal $\frac{1}{2} r^2 (\frac{\pi}{2} - 1)$. Thus $\text{Area}(E) = \text{Area}(F)$.

(Proved)